HHH-V-002178(Army-GL)
March 31, 1983

INTERIM FEDERAL SPECIFICATION

VEGETABLES, MIXED, THERMOSTABILIZED, TRAY PACK

This interim Federal specification was developed by the US Army Natick Research and Development Laboratories, Natick, MA 01760, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above.

1. SCOPE

1.1 Scope. This specification covers mixed vegetables thermostabilized in tray pack cans for use by the Department of Defense as a component of operational rations.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

TT-C-495	-	Coatings, Exterior, for Tinned Food Cans
PPP-B-636	-	Boxes, Shipping, Fiberboard

(Activities outside the Federal Government may obtain copies of Federal specifications, standards, and commercial item descriptions as outlined under General Information in the Index of Federal Specifications, Standards and Commercial Item Descriptions. The Index, which includes cumulative bimonthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification, and other Federal specifications, and commercial item descriptions required by activities outside the Federal Government for bidding purposes are available without charge from General Services Administration Business Service Centers in Boston, MA; New York, NY; Philadelphia, PA; Washington, DC; Atlanta, GA; Chicago, IL; Kansas City, MO; Fort Worth, TX; Houston, TX; Denver, CO; San Francisco, CA; Los Angeles, CA; and Seattle, WA.

FSC 8915

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(Federal Government activities may obtain copies of Federal standardization documents, and the Index of Federal Specifications, Standards and Commercial Item Descriptions from established distribution points in their agencies.)

Military Specifications:

MIL-L-1497 - Labeling of Metal Cans for Subsistence Items
MIL-L-35078 - Loads, Unit; Preparation of Nonperishable Subsistence Items; General Specification for

Military Standards:

MIL-STD-129 - Marking for Shipment and Storage

(Copies of Military specifications and standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

Laws and Regulations:

U.S. Department of Health and Human Services

Federal Food, Drug, and Cosmetic Act and Regulations Promulgated Thereunder (21 CFR Parts 1-199)

(Application for copies should be addressed to the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.)

U.S. Department of Agriculture (USDA)

Regulations Governing Inspection and Certification of Processed Fruits and Vegetables and Related Products

(Application for copies should be addressed to the Director, Fruit and Vegetable Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20230.)

U.S. Standards for Condition of Food Containers

(Application for copies should be addressed to the Director, Food Quality Assurance Division, Agricultural Marketing Service, U.S. Department of Agriculture, Washington, DC 20250.)

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3. REQUIREMENTS

3.1 First article. When specified, a sample shall be subjected to first article inspection (see 4.3, 6.1 and 6.4).

3.2 Ingredients. All ingredients shall be clean, sound, wholesome and free from foreign material, evidence of rodent or insect infestation, harmful extraneous material, off-flavors, off-odors, and off-colors.

3.2.1 Vegetables. The vegetables required for the mixture shall be fresh or frozen green beans, carrots, yellow corn, sweet-type peas and green lima beans. Each of the vegetables shall be of a single varietal type that is suitable for canning.

3.2.2 Salt. Salt used for making the brine shall be non-iodized, sodium chloride having a minimum purity of 99.9 percent.

3.3 Vegetable mixture preparation. Each of the vegetables shall be properly prepared and blanched and shall be in the following forms and percentages.

Vegetable	Form	Percent by weight in mixture
Green beans	Any one nominal size (sieve size 3 or sieve size 4) cut 1/2 to 1-1/4 inches	20 to 30
Carrots	Nominal 3/8 to 1/2-inch dice	20 to 30
Corn	Whole kernel	20 to 30
Peas	Shelled	10 to 20
Lima beans	Shelled	10 to 20

3.4 Brine preparation. The brine shall be made up at the rate of 12 pounds of sodium chloride per 100 gallons of water.

3.5 Tray pack filling and sealing. Each tray pack can (see 5.1.1) shall be filled with prepared vegetable mixture and sufficient brine such as to conform to the following requirements:

a. The average net weight shall be 104 ounces minimum.
b. No individual can shall contain less than 102 ounces of product.
c. No individual can shall contain less than 70 ounces drained weight of product.
d. The cans shall be hermetically sealed immediately after filling.

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3.6 Tray pack thermoprocessing. The filled and sealed tray pack cans shall be thermostabilized by retorting until a sterilization value (F_o) of not less than 9.6 has been achieved.

3.7 Finished product requirements. The finished product shall comply with the following requirements:

a. Vegetables in the finished product shall possess similar varietal characteristics.
b. Texture of the vegetables shall not be hard or mushy.
c. Vegetables in the finished product shall possess a natural cooked color.
d. Vegetables shall be free from extraneous material and defects (e.g. damaged unit).
e. The finished product shall be equal to or better than the approved preproduction sample (see 6.1) in palatability and overall appearance.

3.8 Plant qualification. The product shall be prepared, processed, and packaged under sanitary conditions in accordance with good manufacturing practice. The product shall be processed in establishments meeting the sanitary requirements of the applicable government inspection agency.

3.9 Federal Food, Drug, and Cosmetic Act. All deliveries shall conform in every respect to the provisions of the Federal Food, Drug, and Cosmetic Act and regulations promulgated thereunder.

4. QUALITY ASSURANCE PROVISIONS

4.1 Contractor's responsibility. Inspection and acceptance by the USDA shall not relieve the contractor of obligation and responsibility to deliver a product complying with all requirements of this specification. The contractor shall assure product compliance prior to submitting it to the USDA for any inspection.

4.2 Inspection and acceptance service. Product acceptability shall be determined by the USDA. The USDA will determine the degree of acceptance service necessary to assure compliance with the requirements of this specification. The cost of grading and acceptance services performed by the USDA involving inspection, official documentation, and related services shall be borne by the contractor.

4.3 First article inspection. When a first article is required (see 6.1), it shall be inspected in accordance with the quality assurance provisions of this document and evaluated for overall appearance and palatability. Any failure to conform to the quality assurance provisions of this document or any appearance or palatability failure shall be cause for rejection of the first article.

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4.4 Component and material inspection. Components and materials shall be inspected in accordance with all the requirements of referenced specifications, drawings, and standards unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.1 Ingredient examination. Conformance of ingredients to identity and condition requirements shall be determined by examination of pertinent labels, markings, US Grade certificates, invoices, certificates of analysis, or other valid documents. In addition, each ingredient shall be examined organoleptically as necessary to determine conformance to the condition requirements. Any nonconformance to an identity, condition, or other requirement shall be cause for rejection of the ingredient lot or of any involved product.

4.5 In-process examination. In-process examination shall be performed to determine conformance to the preparation, processing, can interior coating, filling, sealing, and packing requirements. Any nonconformance revealed by actual examination or by review of records of time, temperature, and formulation or other valid documents shall be cause for rejection of the involved product.

4.6 Tray pack inspection. The inspection lot shall include only tray packs produced in one workshift. The USDA reserves the right to separate the inspection lot into smaller inspection lots.

4.6.1 Net weight inspection. Randomly select 30 filled and sealed tray pack cans from the inspection lot and weigh separately. Subtract the average tare weight (determined by randomly selecting and weighing 30 of the empty tray pack cans and lids used in preparing the product and dividing by 30) from the weight of each tray pack in the sample. The results shall be reported to the nearest 0.1 ounce. If the average net weight is less than 104 ounces or if the net weight in any individual can is less than 102 ounces, the lot shall be rejected.

4.6.2 Product inspection. The tray packs shall be inspected for the defects listed in table I. The sampling and acceptance criteria shall be in accordance with the Regulations Governing Inspection and Certification of Processed Fruits and Vegetables, Processed Products Thereof, and Certain Other Processed Food Products.

TABLE I. Product inspection 1/ 2/

Category	Defect
Major	
151	Drained weight of vegetables in a can is less than 70 ounces. 3/
152	Any vegetable not possessing similar varietal characteristics.

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TABLE I. Product inspection 1/ 2/ (cont'd)

Category	Defect
Major	
153	Any vegetable not possessing a natural cooked color.
154	Texture of vegetable mixture hard or mushy.
155	Vegetables not proportionally mixed.
156	Presence of vegetables containing extraneous material or defects (e.g. damaged unit).

1/ The presence of any foreign material (e.g. glass, dirt, insect parts, hair, wood, metal, etc.), foreign odor or flavor (e.g. burnt, scorched, moldy, rancid, sour, stale, etc.) or foreign color shall be cause for rejection of the lot.

2/ Product not equal to or better than the approved preproduction sample in (see 6.1) in palatability or overall appearance shall be cause for rejection of the lot. (This comparison shall be performed only when deemed necessary by the AMS agent).

3/ To determine drained weight, the free liquid in the can shall be poured off and the remaining contents shall be poured into a US Standard No. 8 inch sieve in a manner that will distribute the product over the sieve without breaking the vegetable pieces. Sieve area shall be such that the distributed product does not completely cover all the openings of the sieve. The sieve shall be tilted slightly and allowed to drain for 2 minutes before determining the drained weight by subtracting the sieve tare weight from the gross weight. The drained weight shall be reported to the nearest 0.1 ounce.

4.7 Can condition examination. Examination of filled and sealed tray pack cans shall be in accordance with the United States Standards for Condition of Food Containers, except that labeling shall be inspected in accordance with MIL-L-1497 (see 5.4).

4.8 Shipping container examination. When shipping containers are required to be in accordance with PPP-B-636, examination for defects in construction, closure, and reinforcement shall be in accordance with the appendix of PPP-B-636. In addition, the following defects shall be classified as follows: Major: National stock number, item description, contract number, or date of pack markings missing, incorrect, or illegible. Minor: (1) Other required markings missing, incorrect, or illegible; (2) Interior packing with fiberboard liner or pads not as specified; (3) When applicable, reinforcement with nonmetallic strapping or tape is not used.

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4.9 Unit load inspection. Inspection of unit loads shall be in accordance with the quality assurance provisions of MIL-L-35078.

5. PREPARATION FOR DELIVERY

5.1 Packaging. The product shall be packaged in accordance with level A or C as specified (see 6.1).

5.1.1 Level A. A size 1001 by 1206 by 200 tray pack can shall be filled with 104 ounces of food product, sealed, and thermoprocessed as specified in 3.5 and 3.6. The base plate of the can shall be electrolytic chromium-coated steel and shall be of sufficient weight and temper to protect the product during shipment and storage. The entire inside area of the can shall be coated. The can shall be coated overall on the outside with a coating conforming to type I of TT-C-495.

5.1.2 Level C The product shall be packaged as specified in 5.1.1, except that cans with commercial exterior coating will be acceptable.

5.2 Packing. The product shall be packed in accordance with level A, B or C as specified (see 6.1).

5.2.1 Level A packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed, closed and reinforced in accordance with style RSC-L, grade V2s of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The pads shall be placed between the cans and on the top and bottom of the stacked cans. The pads shall be full length and width less approximately 1/4 inch in each dimension. The box liner and pads shall be fabricated of grade V3c fiberboard.

5.2.2 Level B packing. Four cans of product, packaged as specified in 5.1, shall be packed as specified in 5.2.1, except the box shall be constructed of grade V3c, V3s, or V4s fiberboard.

5.2.2.1 Nonmetallic reinforcement of shipping containers. When specified (see 6.1 and 6.3), the shipping container specified in 5.2.2 shall be reinforced with nonmetallic strapping or pressure-sensitive adhesive, filament reinforced tape in accordance with the appendix of PPP-B-636.

5.2.3 Level C packing. Four cans of product, packaged as specified in 5.1, shall be packed in a snug-fitting fiberboard box, constructed and closed in accordance with style RSC-L, class domestic, grade 275 of PPP-B-636. The cans shall be packed flat, four in depth within the box. The inside of each box shall be provided with a box liner and five fiberboard pads. The pads shall be full length and width less approximately 1/4 inch in each dimension and shall be fabricated of class domestic fiberboard.

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5.3 Unit loads. When specified (see 6.1), the product, packed as specified in 5.2, shall be arranged in unit loads in accordance with MIL-L-35078 for the type and class of load specified. For distribution to the Navy and Air Force, when unit loads are strapped, the strapping shall be limited to nonmetallic strapping.

5.4 Labeling. Each tray pack can shall be labeled in accordance with MIL-L-1497. Paper labels are not permitted. Cans shall show the following statements:

TO HEAT

a. To heat in water: Submerge unopened can in boiling water. Simmer gently 15 minutes. Remove from water and open lid for serving.

b. To heat in oven: Either punch several holes in lid of can or open can in usual manner leaving the loose lid in place to prevent moisture loss. Place in a 350°F oven for approximately 15 minutes until contents reach an internal temperature of 165°F.
WARNING: Do not place closed can in oven as heating will increase internal pressure which may cause the can to burst.

YIELD

Approximately 25 servings of 4 ounces each.

5.5 Marking.

5.5.1 Shipping containers. In addition to any special marking required by the contract, shipping containers shall be marked in accordance with MIL-STD-129.

5.5.2 Unit loads. Unit loads shall be marked in accordance with MIL-L-35078.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) When a first article is required (see 3.1, 4.3 and 6.4).
(c) Provisions for approved preproduction samples (see 3.7 and 6.4).
(d) Level of packaging and packing required (see 5.1 and 5.2).
(e) When nonmetallic reinforcement is required for level B packing (see 5.2.2.1).
(f) Type and class of unit load when unit loading is required (see 5.3).

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6.2 Appropriate level of pack. Based on the conditions known or expected to be encountered during shipment, handling and storage of the specific item being procured, the procuring activity should select the appropriate level of pack in accordance with the criteria established in AR 700-15/NAVSUPINST 4030.28/AFR 71-6/MCO 4030.33A/DLAR 4145.7.

6.3 Selection of nonmetallic reinforcement. The nonmetallic reinforcement specified in 5.2.2.1 should be selected for shipping containers that are destined for transfer at sea or other specific overseas operations.

6.4 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of DAR 7-104.55. The first article should be a preproduction sample. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

Preparing activity:

Army - GL

Project No. 8915-A665

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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